12TH BIOLOGY · CHAPTER 20 · CHAPTER TEST

Biotechnology

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Biology	**Total Marks:** 26
Chapter: 20	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Which bacterium is used in the production of insulin by genetic engineering?

(a) Saccharomyces (b) Rhizobium
(c) Mycobacterium (d) Escherichia

2. What is the use of plasmid in genetic engineering?

(a) To cut DNA (b) To carry DNA to a cell
(c) To join pieces of DNA (d) To replicate DNA

3. In PCR, the optimum temperature for the extension of primer is:

(a) 72°C (b) 98°C
(c) 60°C (d) 37°C

4. Restriction endonucleases are also called:

(a) Molecular glue (b) Molecular carriers
(c) Molecular scissors (d) Molecular probes

5. In blue-white screening, the recombinant colonies are:

(a) Blue (b) White
(c) Red (d) Green

6. Insect-resistant plants carry genes of:

(a) Escherichia coli (b) Thermus aquaticus
(c) Bacillus thuringiensis (d) Aspergillus

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Define genetic engineering.
ii. What is the role of vectors in recombinant DNA technology?
iii. What are sticky ends?
iv. What is Taq polymerase? Why is it used in PCR?
v. Differentiate between endonuclease and ligase.
vi. What is golden rice?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe the steps of recombinant DNA technology.
(b) Explain the formation of human insulin protein in bacteria.

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